

DL Experiment Synthesis Report

4DVarNet-FM: DL Experiment Report

Experiment configurations:

ID	Model	Channels	Epochs	Train Mix	Params
B1_small_unet	Tweedie	[32, 64, 128]	200+400	cs1+cs2	fixed
C1_longer_train	Tweedie	[64, 128, 256]	400+800	cs1+cs2	fixed
C4_stage1_only	Tweedie	[64, 128, 256]	200	cs1+cs2	fixed
D1_cs2_only	Tweedie	[64, 128, 256]	200+400	cs2_only	fixed

Test data:

200 trajectories per case study (seed=123 CS1, seed=124 CS2)
T_max = 3.0s dt = 0.01 15 obs per window (obs_interval=20)

Reference (B1_small_unet): CS1 $\mu=0.0973$ CS2 $\mu=0.1104$ Deg=1.13x

DL Experiments — Metrics

Summary of Results – Mean RMSE per Component

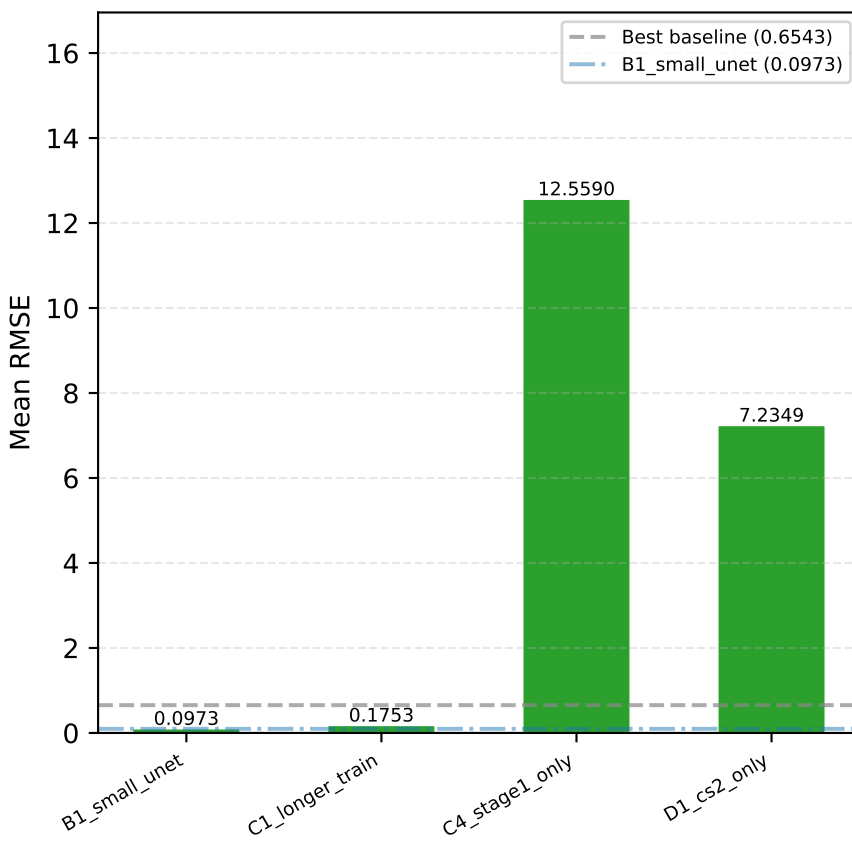
ID	Model	CS1 X	CS1 Y	CS1 Z	CS1 μ	CS2 X	CS2 Y	CS2 Z	CS2 μ	Deg	
B1_small_unet	Tweedie	0.0811	0.0979	0.1129	0.0973	0.0998	0.1090	0.1222	0.1104	1.13	x
C1_longer_train	Tweedie	0.1715	0.1647	0.1896	0.1753	0.1782	0.1660	0.1924	0.1788	1.02	x
C4_stage1_only	Tweedie	5.2387	9.3100	23.1282	12.5590	5.3403	9.3120	22.5582	12.4035	0.99	x
D1_cs2_only	Tweedie	7.7537	6.5104	7.4407	7.2349	7.2473	6.6061	7.5978	7.1504	0.99	x

Best baseline reference
(best baseline CS1 μ=0.6543, CS2 μ=1.5497)

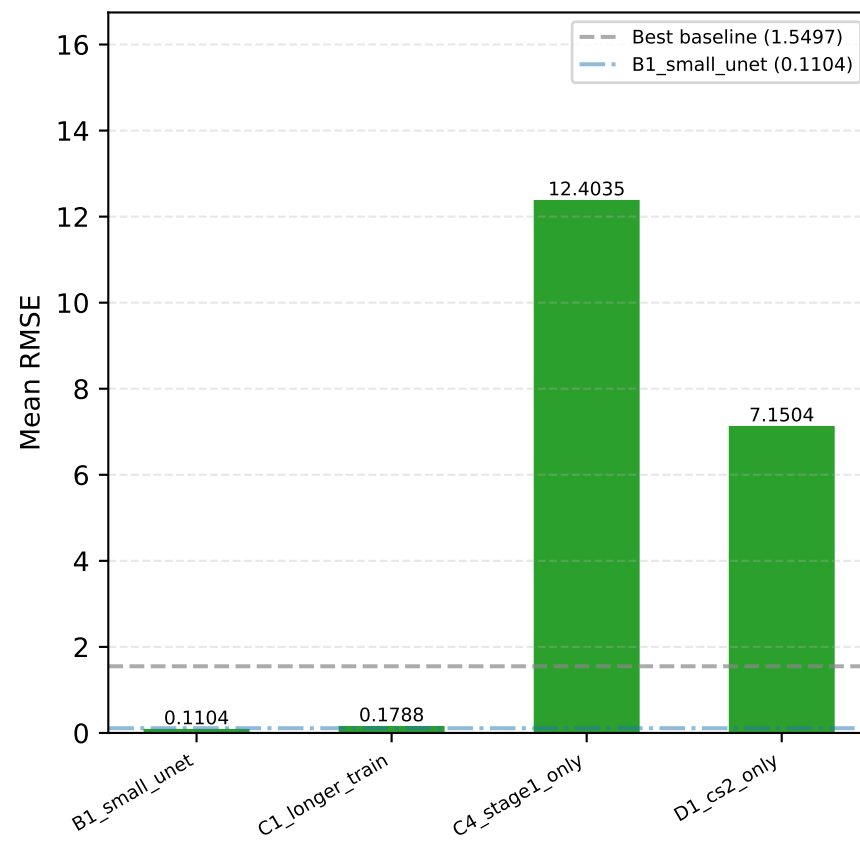
Degradation = CS2 μ / CS1 μ. Lower = more robust.

DL Experiment Comparison — Mean RMSE

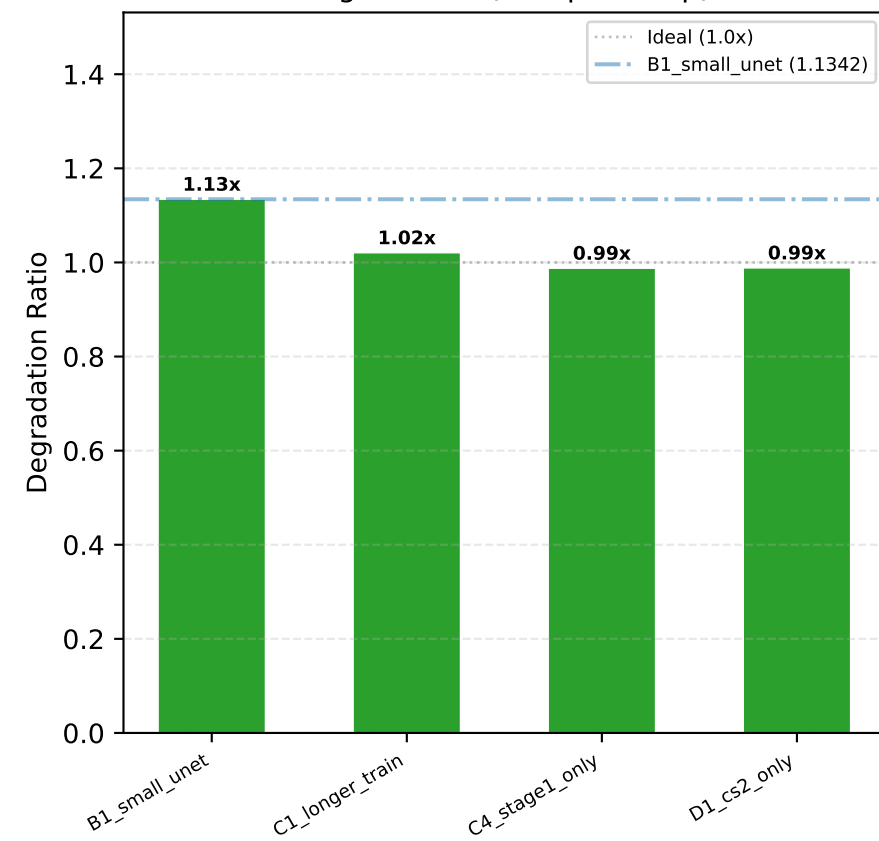
CS1 Mean RMSE



CS2 Mean RMSE



Degradation (CS2 μ / CS1 μ)



B1_small_unet — Trajectories (no data)

No trajectory data for B1_small_unet

C1_longer_train — Trajectories (no data)

No trajectory data for C1_longer_train

C4_stage1_only — Trajectories (no data)

No trajectory data for C4_stage1_only

D1_cs2_only — Trajectories (no data)

No trajectory data for D1_cs2_only